# NEEVETHA N

**Software Systems Student | Full-Stack & AI Systems Developer**

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## PROFESSIONAL SUMMARY

Software Systems student (CGPA 8.89/10) with strong engineering foundations in full-stack web development, distributed microservices, and AI-enabled application architecture. Experienced in building production-grade web systems - from real-time WebSocket listening platforms to sandboxed code execution environments - demonstrated through national hackathon wins, research internships, and scalable full-stack projects.

## EDUCATION

**Coimbatore Institute of Technology (CIT)** - M.Sc. Software Systems (5-Year Integrated) 2024 - 2029 | Coimbatore, India
**Academic Standing:** CGPA 8.89 / 10.0 | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Web Technologies

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript, Python, SQL, C++, C, Java

**Frontend Development:** React.js, Next.js 14 (App Router), Tailwind CSS, Zustand, TanStack Query, D3.js, Recharts, HTML5, CSS3

**Backend & APIs:** FastAPI, Node.js, Express, RESTful APIs, WebSockets, Celery, Pydantic, Microservices Architecture

**Databases & Caching:** PostgreSQL, Supabase, Redis, SQLite, MySQL, SQLAlchemy, Alembic

**DevOps & Engineering Tools:** Docker, Docker Compose, Nginx, Git, GitHub, Postman, Linux / Bash

## FEATURED ENGINEERING PROJECTS

**PulseBoard** - *Real-Time Social Trend & Brand Risk Analytics Platform* [github.com/niviii-08/PulseBoard]

**Tech Stack:** React, TypeScript, FastAPI, Celery, Redis, PostgreSQL, SQLAlchemy, Alembic, WebSockets, Docker, Nginx, D3.js

- **Real-Time Streaming Architecture:** Architected an end-to-end social monitoring platform delivering live conversation momentum and risk analytics to a React dashboard via bidirectional WebSockets.
- **Asynchronous Ingestion Pipeline:** Built distributed background workers using FastAPI, Celery, and Redis to ingest multi-source web signals without blocking client queries.
- **Data Modeling & Containerization:** Designed relational schemas and migrations with PostgreSQL, SQLAlchemy, and Alembic; containerized frontend, API, broker, and reverse proxy using Docker and Nginx.

**ContextIQ** - *Personal Behaviour Intelligence & Cognitive Friction Reduction Platform*

**Tech Stack:** Next.js 14, React, TypeScript, FastAPI (5 Microservices), PostgreSQL, SQLAlchemy, Docker Compose, Tailwind CSS

- **Modular Microservices Architecture:** Engineered a scalable system decoupling operations across 5 FastAPI microservices (Core API, Data Engine, Context Engine, ML Engine, Intelligence Service).
- **Next.js 14 App Router Frontend:** Developed a responsive dashboard in TypeScript with Tailwind CSS, utilizing client-side state stores and optimistic UI updates for real-time interaction.
- **Multi-Container Orchestration:** Orchestrated multi-service backend containers and automated PostgreSQL database provisioning using Docker Compose with strict Pydantic contract validation.

**LifeLink** - *Intelligent Blood & Emergency Response System* [github.com/niviii-08/Lifelink]

**Tech Stack:** React, TypeScript, Node.js, Express, Python, FastAPI, Firebase Realtime Database, SQLite, Leaflet Maps

- **Dual-Backend Engineering:** Engineered a hybrid backend leveraging Node.js/Express for real-time dispatch events and Python/FastAPI for donor-patient compatibility algorithms.
- **Live Geospatial Tracking:** Integrated Firebase Realtime Database for live status broadcasts, paired with an interactive Leaflet.js mapping interface for regional blood inventory tracking.
- **Responsive Workflow Design:** Built mobile-first interfaces in React & TypeScript, enabling rapid multi-step emergency blood requests and appointment scheduling.

**SkillSync** - *AI-Powered Skill Assessment & Hackathon Platform* [github.com/niviii-08/SkillSync]

**Tech Stack:** Next.js 14, TypeScript, Zustand, Supabase (Auth & Database), Piston Code Execution Engine, Tailwind CSS, GPT-4o

- **Sandboxed Code Execution Engine:** Integrated the Piston remote execution engine to safely compile and run user-submitted code in isolated environments with stdout/stderr capture.
- **Authentication & Cloud Database:** Configured Supabase Auth with secure session handling and Row-Level Security (RLS) policies on a PostgreSQL backend.

- **Complex Client State Management:** Managed multi-step test workflows, webcam proctoring, and live coding state transitions using Zustand state stores in Next.js 14.

## PROFESSIONAL EXPERIENCE

**Funobotz Paper Robotics Pvt. Ltd.** - Online Research Intern (via Praya Labs) May 2026 - Jun 2026 | Remote

*Credential: Certificate No. INPLNE026 | Focus: Conversational AI & Educational Chatbot Systems*

- **Conversational Logic Architecture:** Contributed to the core conversational dialogue trees and intent routing logic of the Funobotz AI Chatbot System across the development lifecycle.
- **Validation & Quality Assurance:** Collaborated with engineering leads on automated API testing, dialogue edge-case verification, and system validation to ensure high intent classification reliability.

## HONORS & HACKATHON ACHIEVEMENTS

- **RAC-A-THON 2025:** Awarded Best Theme & Special Mention in a 24-hour national-level hackathon organized by Rotaract Club.
- **Codeathon 2.0 (Thiran 2026):** Selected for an Internship Opportunity through competitive project evaluation at the national-level 24-hour hackathon.
- **Tech Sprint 2025:** Recognized with Special Mention for innovative architectural solution design.
- **Technical Competitions:** Active participant and finalist at Melinia 2026 and Intellina 2026 technical symposiums.

## CERTIFICATIONS & LEADERSHIP

**Industry Certifications:** Career Essentials in Generative AI (Microsoft & LinkedIn Learning, 2024) | Data Analysis Job Simulation (Deloitte) | MVP to AI-Powered Apps for Modern Web Developers

**Leadership & Campus Roles:** Video Editing Lead, Google Developer Groups (GDG), CIT (2025-2026) | Event Chairperson, Rotaract Club, CIT (2026)